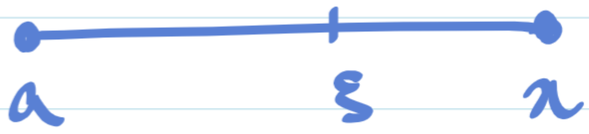
13.6.2 Linear Approximation. 22/10/2026

Note: For $y=f(x)$, using Taylor Formula. (MVT)

$$f(x) = f(a) + f'(a)(x-a) + R$$



$$R = \frac{f''(\xi)(x-a)^2}{2!}\,;$$ ξ is bet x and a

assume $x \simeq a$, $x-a \simeq 0$

$$\Rightarrow f(x) \simeq \underbrace{f(a) + f'(a)\cdot(x-a)}$$

L = linear approx for f (n=1)

$$\text{Error} = |R| \le M\frac{(x-a)^2}{2} \qquad (\text{when } |f''(\,)| \le M)$$

Ref to Defn [p 784] - Sect 13.3

$$f\begin{bmatrix} a+\Delta x \\ b+\Delta y \end{bmatrix} = f\begin{bmatrix} a \\ b \end{bmatrix} + \underset{\sim}{\nabla} f\begin{bmatrix} a \\ b \end{bmatrix}\cdot[\Delta x \ \Delta y] + \underset{\sim}{\varepsilon}\cdot[\Delta x \ \Delta y]$$

when $\Delta x \to 0$, $\Delta y \to 0$, $\underset{\sim}{\varepsilon} = [\varepsilon_1 \ \varepsilon_2] \to [0 \ 0]$

$$\Rightarrow f\begin{bmatrix} a+\Delta x \\ b+\Delta y \end{bmatrix} \simeq \underbrace{f\begin{bmatrix} a \\ b \end{bmatrix} + \underset{\sim}{\nabla} f\begin{bmatrix} a \\ b \end{bmatrix}\cdot[\Delta x \ \Delta y]}$$

L = linear Approx or linearization of f at $\begin{bmatrix} a \\ b \end{bmatrix}$

write $x = a+\Delta x$

$y = b+\Delta y$

a) Linearization

$$f\begin{bmatrix} x \\ y \end{bmatrix} \simeq L = f\begin{bmatrix} a \\ b \end{bmatrix} + f_x\begin{bmatrix} a \\ b \end{bmatrix}(x-a) + f_y\begin{bmatrix} a \\ b \end{bmatrix}(y-b)$$

b) Error Term

$$\text{Err.} \le \frac{(|x-a| + |y-b|)^2}{2} M$$

We can select $M \ge |f_{xx}|, |f_{xy}|, |f_{yy}|$

Note:

We can extend the idea to the case $n=3$

namely, for $w = f(x\ y\ z)$

a) Linearisation

$$L = f\begin{bmatrix} a \\ b \\ c \end{bmatrix} + f_x\begin{bmatrix} a \\ b \\ c \end{bmatrix}(x-a) + f_y\begin{bmatrix} a \\ b \\ c \end{bmatrix}(y-b) + f_z\begin{bmatrix} a \\ b \\ c \end{bmatrix}(z-c)$$

b) Error Term

$$|Er| \leq \frac{M}{2}\left(|x-a| + |y-b| + |z-c|\right)^2$$

Here $M \geq |f_{xx}|, |f_{xy}|, |f_{xz}|$

$|f_{yx}|, \ldots\ldots$

$\ldots\ldots |f_{zz}|$

Example. Find the linearization $L(x,y,z)$ of the function at P_0.

13.6.50. $f(x,y,z) = \sqrt{2}\cos x \sin(y+z)s$ at $P_0(0,0,\pi/4)$,

R: $\quad |x| \le 0.01,\ |y| \le 0.01,\ |z-\pi/4| \le 0.01$

$a=0 \quad b=0 \quad c=\frac{\pi}{4}$

$|\Delta x| \le |x-a| \le 0.01$

$|\Delta y| \le |y-b| \le 0.01$

$|\Delta z| \le |z-c| \le 0.01$

Soln

$f = \sqrt{2}\cos x \sin(y+z) \qquad f(P_0) = \sqrt{2}\frac{1}{\sqrt{2}} = 1$

$f_x = -\sqrt{2}\sin x \sin(y+z) \qquad f_x(P_0) = 0$

$f_y = \sqrt{2}\cos x \cos(y+z) \qquad f_y(P_0) = \sqrt{2}\cdot 1 \cdot \frac{1}{\sqrt{2}} = 1$

$f_z = \sqrt{2}\cos x \cos(y+z) \qquad f_z(P_0) = 1$

Hence the linearization of f at P_0

$$L = 1 + 0(x-0) + 1(y-0) + 1\left(z-\frac{\pi}{4}\right)$$

$$= 1 + y + z - \frac{\pi}{4}$$

Extra Question: Find a max error term.

[Cont.] Extra Question: Find a max error term.

$$f_x = -\sqrt{2}\sin x \sin(y+z)$$

$$f_y = \sqrt{2}\cos x \cos(y+z)$$

$$f_z = \sqrt{2}\cos x \cos(y+z)$$

$$f_{xx} = f_{yy} = f_{zz} = -\sqrt{2}\cos x \sin(y+z)$$

$$f_{xy} = f_{xz} = -\sqrt{2}\sin x \cos(y+z)$$

$$f_{yz} = -\sqrt{2}\cos x \sin(y+z)$$

Since $|\sin(\ )| \le 1$ and $|\cos(\ )| \le 1$

In this case, we take $M = \sqrt{2}$

$\therefore$ max error term is

$$= \frac{1}{2} M \left(|\Delta x| + |\Delta y| + |\Delta z|\right)^2$$

$$= \frac{\sqrt{2}}{2}(0.03)^2 = \text{——} \quad \#$$

13.6.3 Differentials

Definition (pp. 811)**.** Let $w = f(x, y, z, \cdots)$ be a differential function. The *(total) differential* of f is given by

$$(dw \text{ or }) \; df = f_x dx + f_y dy + f_z dz + \cdots .$$

Comment 13.6.1.

1. dw gives an approximation for the change in w wrt. the changes Δx, Δy, $\cdots$ in x, y, $\cdots$ respectively.
2. Value(s) of x or y or w INCREASED $\Rightarrow$ sign for dx or dy or dw is "+".
3. Value(s) of x or y or w DECREASED $\Rightarrow$ sign for dx or dy or dw is "−".

Example (13.6.52). The resistance R produced by wiring resistors of R_1 and R_2 ohms in parallel (see accompanying in pp. 815) can be calculated from the formula

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}.$$

(a) Show that $dR = \left(\frac{R}{R_1}\right)^2 dR_1 + \left(\frac{R}{R_2}\right)^2 dR_2$.

(b) You have designed a two-resistor circuit, like the one shown, to have resistances of $R_1 = 100$ ohms and $R_2 = 400$ ohms, but there is always some variation in manufacturing and the resistors received by your firm will probably not have these exact values. Will the value of R be more sensitive to variation in R_1 or to variation in R_2? Give reasons for your answer.

(c) In another circuit like the one shown, you plan to change R_1 from 20 to 20.1 ohms and R_2 from 25 to 24.9 ohms. By about what percentage will this change R?

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